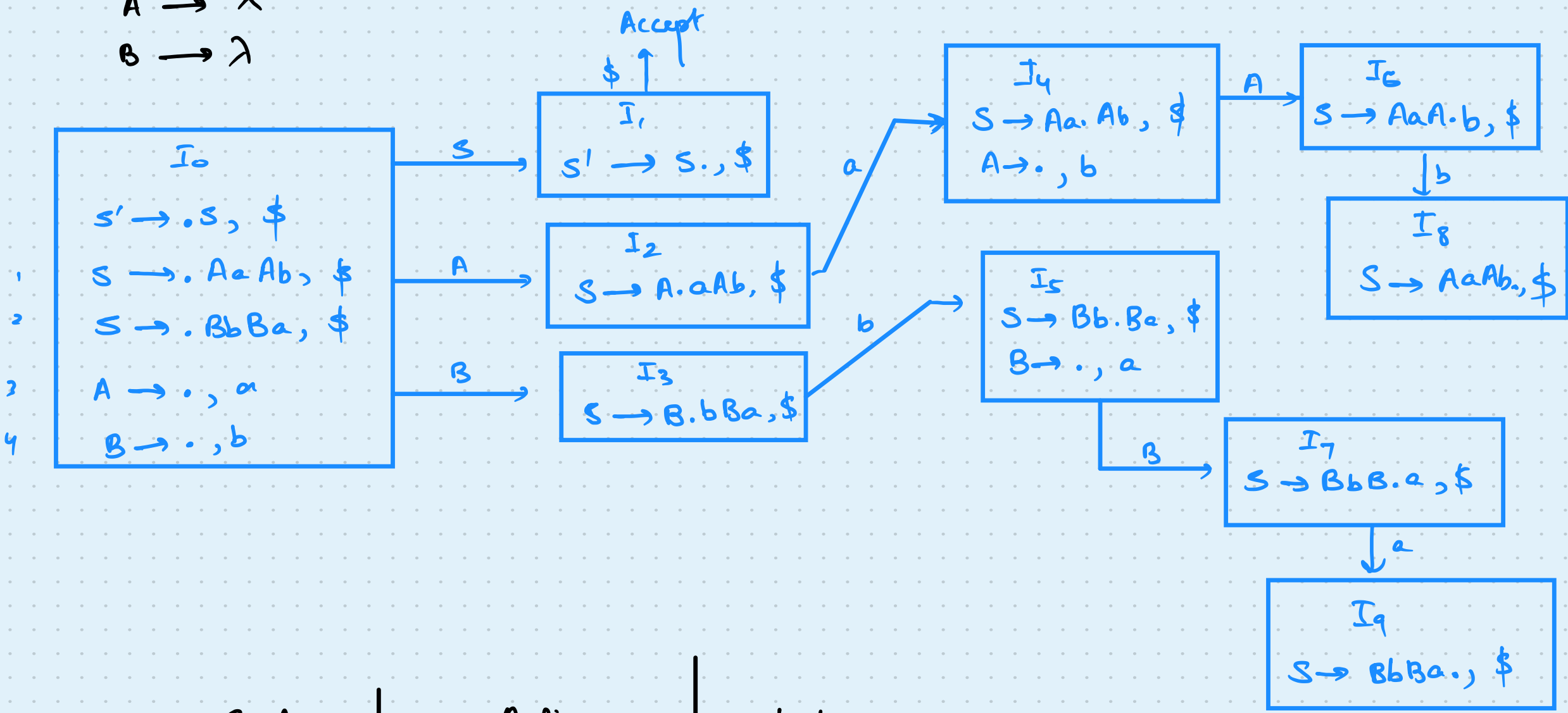


LR(1) - Continuation

Eg: $S \rightarrow AaAb \mid BbBa$
 $A \rightarrow \lambda$
 $B \rightarrow \lambda$



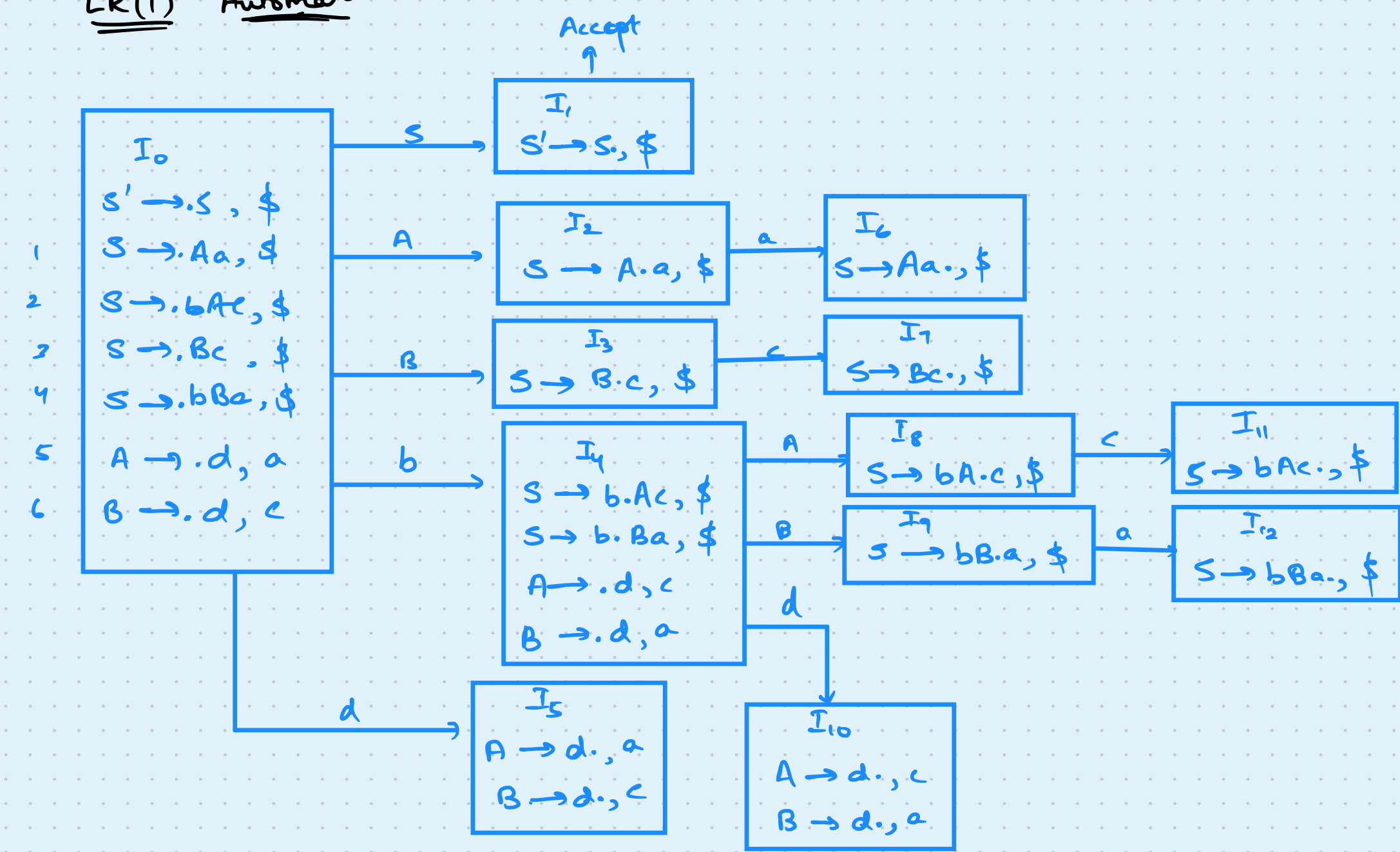
State	Action			Goto		
	a	b	\$	S	A	B
0	R ₃	R ₄		1	2	3
1			Accept			
2	S ₄					
3		S ₅				
4		R ₃		6		
5	R ₄					7
6		S ₈				
7	S ₉					
8			R ₁			
9		R ₂				

LALR(1)

Construct LR(1) and merge states with identical items but different lookahead symbols
Replace individual state numbers with combination of the two

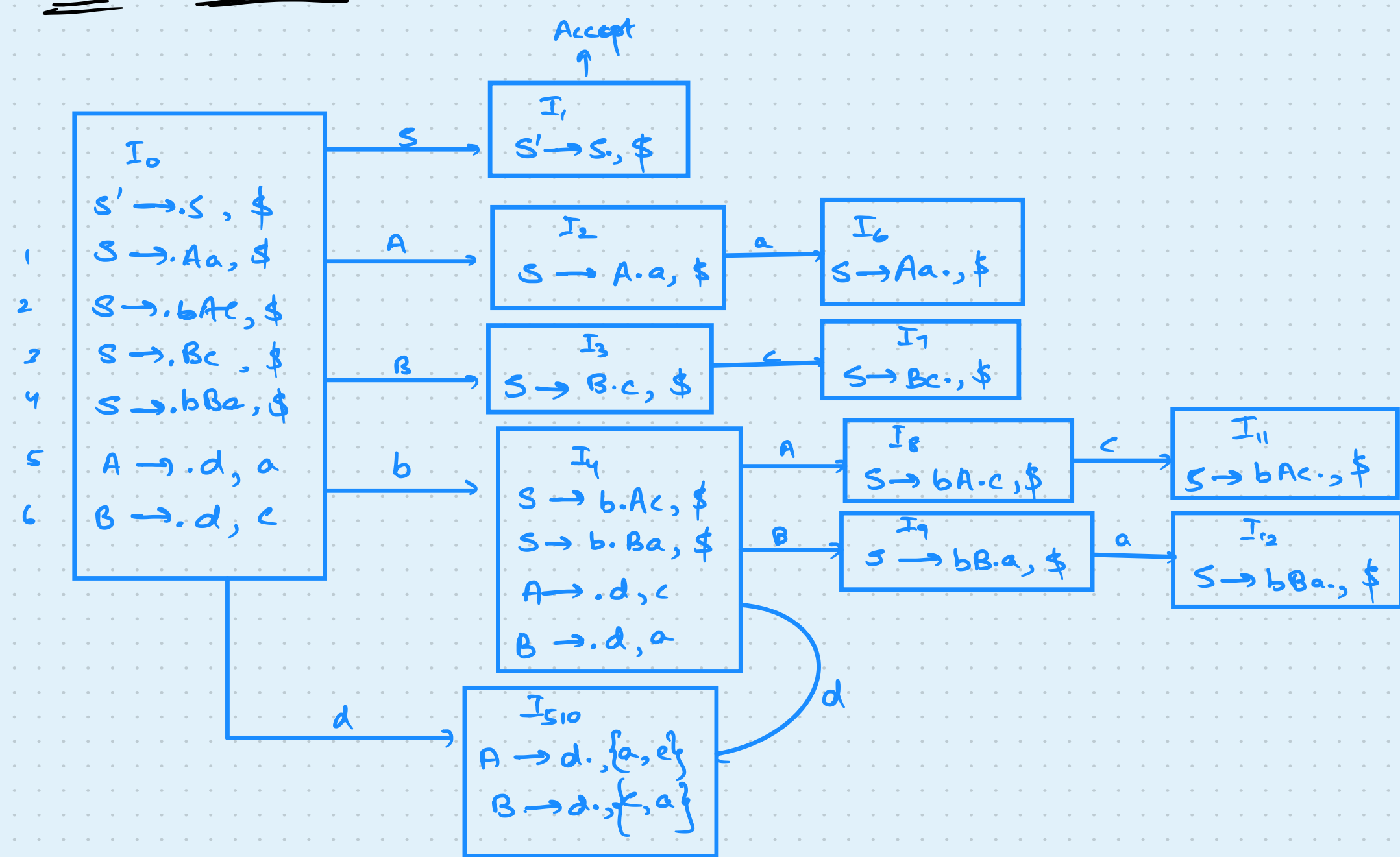
Eg: $S \rightarrow Aa \mid bAc \mid Bc \mid bBa$
 $A \rightarrow d$
 $B \rightarrow d$

LR(1) Automata



State	Action					Goto		
	a	b	c	d	\$	S	A	B
0		S ₄		S ₅		1	2	3
1					Accept			
2	S ₆							
3			S ₇					
4				S ₁₀		8		9
5	S ₅		S ₆					
6					S ₁₁			
7					S ₁₃			
8			S ₁₁					
9	S ₁₂							
10	S ₆		S ₅					
11					S ₁₂			
12					S ₁₄			

LALR automata



State	Action					Goto		
	a	b	c	d	\$	S	A	B
0		S ₄		S ₁₀		1	2	3
1					Accept			
2	S ₆							
3			S ₇					
4				S ₁₀		8		9
5		S ₅ , S ₆		S ₅ , S ₆				
6					S ₁₁			
7					S ₁₃			
8			S ₁₁					
9	S ₁₂							
10					S ₁₂			
11					S ₁₄			
12								

reduces-reduce
conflict;
not LALR(1)

Yacc

Keywords	Description
%left	Identifies tokens that are left-associative w/ others
%nonassoc	" " " " NOT associative " "
%right	" " " " right-associative " "
%start	Identifies a nonterminal name for the start symbol
%token	Identifies the token names that yacc accepts
%type	Identifies datatype of a given token if applicable

→ All tokens on same line: same precedence and associativity

→ lines are in increasing order of precedence

Eg: %left '+' '-'
%left '*' '/'